Local-First Memory for Agents

The kernel stores what an agent learns. Every write crosses the same gate: the injection screen reads the text, the trust label bounds its provenance, and the audit chain records the write without the bytes.

Recall is a fusion of three legs. The lexical leg matches exact terms through a full-text index. The semantic leg embeds the query and ranks vectors by cosine distance. The graph leg walks typed relationships so a two-hop answer can cite the path it walked.

Nothing in the hot path calls a model provider. Embeddings are local, deterministic, and zero-token; the ranking is a pure function of the stored rows. A recall that cannot run offline is a recall that has already failed its own posture.

The Write Gate

An agent write is a proposal until a human approves it. The proposal queue is the only promotion path: no insert may bypass it, and the behavioral pins prove the negative by counting rows after a write.

Provenance rides every row. The source label is a bounded vocabulary, the owner stamp records the principal, and the origin field derives from the label so an agent cannot forge human authorship.

Quarantine is the middle verdict. A flagged chunk is stored but never retrieved by default: it carries no vector, so the ANN top-k cannot surface it. The reviewer sees the verdict without a round trip.

Determinism as a Product Law

Same input bytes, same output bytes. The law applies to extraction, to embedding, to ranking, and to every golden the tests pin. A path that cannot be re-run into the same bytes is a path that cannot be audited, and an unaudited path does not ship.

The fence is the advisory layer around untrusted context. Storage stays verbatim; the strips compose at the boundary, and the sentinel vocabulary has exactly one definition in the tree.